MITS 5300G: Operating System Security

Course Overview

Course Instructor

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Courses:

- Security policies & risk management
- Malware & software security
- **Operating system security**
- Applied cryptography

Research:

- Primary: computer security and human factors
- Other: malware, software security, operating system security, security policies, biometrics, and BCIs

How to contact me

- Come to class discussions and ask questions! Asking questions shows you are engaged and interested.

- If you have questions about the material outside of class hours, there will be a few ways you can get them answered:
  - Q&A discussion board on Canvas (if you have a question, others probably do too!)
  - Visit me during my weekly office hours
  - Canvas message me for an appointment
    - I will strive to reply within 2 business days.

Survey For Office hours

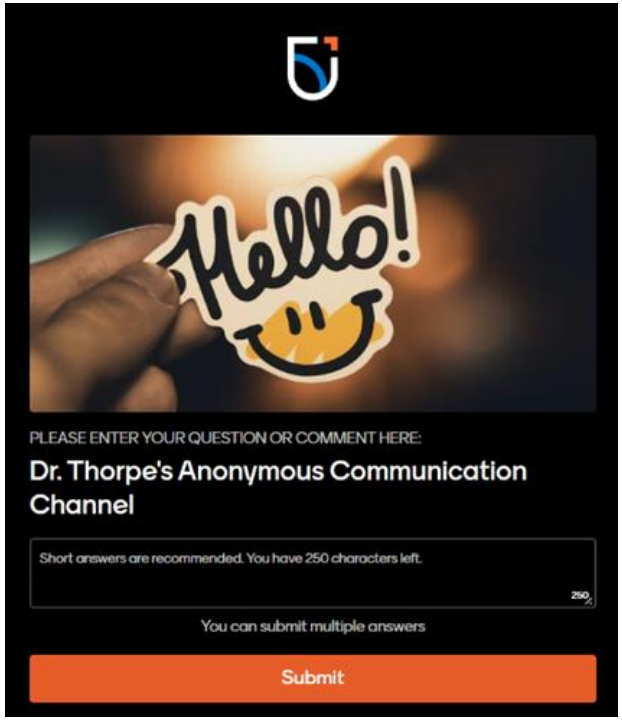
Please go to menti.com with code I will give you

Please select ALL times you can attend

Anonymous Communication With Me

- I like to keep a way for you to let me know about questions, concerns, comments, etc. anonymously.
  - Go to: www.menti.com and use the code I give you.

- If in your comment you are referring to the lecture, please let me know the slide number in question (if relevant).



What are the learning goals of this course?

- Understand the threats to operating systems
- Understand the goals of secure operating systems
- Learn how to use existing operating system security technology and tools
- Learn about selected security features of Linux, Windows, and SELinux
- Understand the challenges and limitations faced in creating secure operating systems
- Apply basic research skills

Course Topics

Let’s look at the Syllabus on Canvas

Textbook and Readings

- Main source of course material is the lecture notes and discussion papers, as posted on Canvas.

- There is a textbook that some of the course notes are based upon (electronic version is freely available from the Ontario Tech library). This is FYI, it is not required reading:
  - T. Jaeger, “Operating System Security (Synthesis Lectures on Information Security, Privacy, and Trust)”, Morgan and Claypool Publishers, edition 1, 2008.

- Readings may be assigned or recommended during the course.

Important Note Regarding Textbook

- Some students have found parts of this book confusing.
  - If you read the book for more information, and find something confusing, do the following:
  1. Read the course notes concerning the topic
  2. If (1) doesn't help, come to my office hours

- Please don't hesitate to ask me questions! I am happy when you ask questions!

What is the “Paper Discussion Lead”?

It means that you thoroughly read and analyze the paper, and your group prepares/delivers:

- A 15-20 minute presentation on it’s material (10 points)
  - Paper #1 can be 7-10 minutes.
- Leads the class discussion of the paper by answering questions (4 points)
- Groups must send PDF of their slides to instructor by midnight the number before their presentation.
- All group members must participate equally!

Sign-up for Discussion Leads

Sign up for a discussion lead through Canvas

- Go to People->“Paper Discussion Leads”

Under Module 1 you will see “Paper discussion lead sign-up”. See the schedule of weekly topics and paper discussions linked in the second sentence.

The group number corresponds to the paper number (e.g., **(1)**) in the schedule

- E.g., if you sign up for “Paper discussion leads 5”, you’ve signed up for the ErsatzPasswords paper.

Note that one paper (#1) has only 1-2 group members, all other papers have up to 4 members in a group.

Paper Discussion Lead (Optional)

If you are interested in finding another paper, that is great, but it must be:

- Proposed to me at least 2 weeks before the date of the presentation.
- Approved by me.
- What will I look for to approve a different paper?
  - Paper appears in a reputable venue (conference, journal, or industry/government whitepaper)
  - Paper is relevant to the topic of that week

Discussion Participation Grading

Top 8 of 12 (of the ones you are not leading).

To get the marks:

- Be present
- Be attentive (i.e., no devices interaction)
- Try to ask questions
- Sign in (method will be announced before or after presentation)

Research Projects: Topics

The project must be on a topic within the theme of operating systems security. Topics must be selected by each student (first-come-first-served) to avoid substantial overlap in class projects. Suitable topics:

- Do not overlap with course topics (including discussion papers)
- Do not overlap with other student's topics (as listed in spreadsheet linked below)
- Are not too general
- Are within the scope of OS security

Research Project: Suitable Topics

1. A literature review of an operating-systems related topic. In this case, the literature review must contain at least 7-10 references from reputable sources.
2. A comprehensive review of the security features and vulnerabilities/threats of a particular operating system not covered in class.
3. A detailed description of the internal workings, algorithms, user interface issues, etc. of a particular standalone tool for OS security enhancement.
4. An evaluation or implementation of a software tool (or proposed software tool) that is relevant to OS security.
5. If you have another idea that you think is in scope, feel free to ask me about it.

Claiming a Research Project Topic

- Go to the “Claimed Project Topics” spreadsheet here: https://docs.google.com/spreadsheets/d/1b5Sjh2bwHXidcWsZLEk9f9A-4wbs7UQVKpMG1qsQ-eU/edit?usp=sharing
- Check the list of existing project topics to ensure there is no overlap with your topic.
- If you are sure your topic doesn’t overlap with someone else’s, you may add your name and topic to the spreadsheet.
- Note: you may contact another student in the class for more clarity on their topic, in order to differentiate your project topic from theirs. It is not acceptable to modify anyone else’s project topic in the spreadsheet.

Research Project: Further Details

- Note it is to be submitted as a **presentation**.
- The presentation should be submitted on Canvas (Research Project Assignment) as a PDF slide deck containing:
  1) Slides used.
  2) On the first/title page, a link to your presentation (which must be to a private YouTube video).
  3) Proper citations (in IEEE format) must be used for all references.

Weekly Quizzes

- Please note you have unlimited attempts to submit each quiz. The latest submission is what will count towards your grade.
- The purpose of these quizzes is not for evaluation, but rather to work as a study aid.
- I recommend trying your best on each attempt to maximize your study benefits (e.g., double-checking your answers before submitting, carefully considering your answers, etc.).

Academic Integrity (1/2)

- Don’t copy from another person.

- Make sure you cite every source of information you used. e.g.,
  - Books, articles, websites,
  - a person you spoke with,
  - images you did not create, and
  - code you did not write

- If an assignment asks you to write something yourself, it is not OK to just cite something. You must write the answer in your own words.

Academic Integrity (2/2)

- Turnitin.com will be used for assignments and projects.
  - This detects any copy-pasting, even at the sentence level.

- Please read the full Ontario Tech policy online.

Academic Integrity and AI

- Please be careful--- some students have gotten into trouble with this in the past.

- OK usage (with extreme caution): to help identify sources that are relevant to your project, then you read those sources directly rather than AI summaries. Sometimes the summaries are inaccurate! You would then write the reports/presentations yourself, citing the sources directly.

- **Not OK to use it to generate the writing for your reports or presentations!**

Academic Integrity and AI (continued)

- If you use AI tools to do grammar-checks and suggest fixes:
  - You must protect yourself by doing the following:
    1. Keep a copy of your first draft (that you wrote yourself), prior to AI-assisted grammar checks.
    2. Keep all other intermediate versions after using the AI tool to improve it.

    - Essentially you are creating a trail that you can then use to prove you wrote it and just used AI assistance for grammar checks.

Break time

If you haven’t signed up for a discussion lead group yet, I encourage you to do so if you have extra time during the break!